been dated by the carbon-14 method, thus placing the beginning of the *Alnus* curve at 6620±125 B.C.[5]. The pollen analysis from the upper part of the mire (Fig. 1,c) indicates that its age is apparently not far from the time when the *Alnus* curve began. This correlates very well with the radiological dating from Sweden.

Marine sediments resting directly upon a boreal terrestrial deposit have apparently not been previously reported in Norway. The occurrence of marine diatoms, as described above, indicates that sea-level was not far below the rock threshold and that waves were able to break over it, for example, during storms and exceptionally high-water levels. The minimum level of the boreal shore-line, judging by the evidence of the diatoms, was not far from the present sea-level in the area. However, work on a graph of the movement of the shore-line is being carried out at present, and more detailed results will be published later.

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Olivine-Spinel Transformation

IN the light of A. E. Ringwood's recent synthesis of a spinel form of fayalite (private communication), Fe_2SiO_4 (the iron end-member of the fayalite–forsterite mixtures which comprise the bulk of natural olivines), it seemed worth while to attempt a synthesis of a spinel form of forsterite, Mg_2SiO_4. Ringwood prepared the fayalite-spinel at about 50 kilobars and 500° C., and his calculations indicated that the forsterite-spinel should become stable at about 100 kilobars and 500° C. It is possible that olivine is abundant in deeper layers of the Earth (50–1,000 km.) and hence some knowledge of its phase transitions would be of geological interest[1].

To study these possible phase transitions, appropriate mixtures of dry and hydrous oxides were loaded into a carbon capsule contained in a high-pressure, high-temperature apparatus. After the pressure on the capsule was increased to the desired level, it was electrically heated to about 550° C. for about 20 min. (Crystals of 10 or 20μ average diameter formed in 5 min.) Then the sample was quenched to room temperature and the pressure was reduced to atmospheric. In this way any phases formed at high pressure would tend to be preserved. Various hydrated phases formed if the maximum temperature did not exceed about 500° C.

Table 1

Charge	Pressure (kilobars)	Products identified
$2\ Mg(OH)_2 + SiO_2 \cdot nH_2O$	95–130	Forsterite
$Fe + Fe_3O_4 + SiO_2 \cdot nH_2O$	110	Ringwood's fayalite-spinel (Debye-Scherrer patterns agree)
$Fe + Fe_3O_4 + MgO + SiO_2 \cdot nH_2O$	110	Forsterite, fayalite-spinel
Hawaiian olivine	93–(hot)	Olivine melted and frozen this run. Olivine recovered
Moist Hawaiian olivine	90–110	Olivine unchanged
$2\ Mg(OH)_2 + 0{\cdot}9\ SiO_2 + 0{\cdot}1\ GeO_2 + H_2O$	110	Forsterite
$2\ Mg(OH)_2 + SiO_2 \cdot nH_2O$ + 5 per cent Ni_2GeO_4	110	Forsterite
$2\ NiO \cdot SiO_2 \cdot nH_2O$	90–110	Nickel forsterite (no spinel)
$2\ ZnO \cdot SiO_2 \cdot nH_2O$	90	New phase, r.i. 1·78, biaxial. No spinel

Table 1 summarizes the results of these experiments.

Thus no other olivinic spinel other than Ringwood's fayalite-spinel formed in these experiments. Perhaps higher pressures are necessary. This work will be continued.

I am indebted to A. E. Ringwood and Prof. F. Birch for stimulating the work on this problem and for providing some of the oxide mixtures.

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PALÆOMAGNETISM

Late Tertiary Geomagnetic Field in North-western Canada

THE fundamental assumption made in interpreting palæomagnetic results is that the average geomagnetic field corresponds to that of a central axial dipole. This communication reports on some palæomagnetic results which support this assumption and which were made on Late Tertiary volcanic rocks, which were collected during the summer of 1955 from north-western Canada by field geologists of the Geological Survey of Canada from flat-lying, undisturbed, vesicular basalts.

Forty-eight samples were collected from four different localities. The largest suite, twenty-nine amples, was collected by J. O. Wheeler from lavas exposed near Whitehorse, Yukon, at a latitude of 60·5° N. and longitude of 135·0° W. Another nine samples were collected by W. H. Poole near Wolf Lake, Yukon, at 60·0° N. and 130·5° W. L. H. Green collected six samples near Selkirk, Yukon, at 63·0° N. and 138·0° W., and J. D. Aitken collected four samples near Surprise Lake, British Columbia, at 59·5° N. and 134·0° W.

These samples were measured on a magnetometer of the rock generator type at the Geological Survey of Canada during 1958. The results of these measurements are shown in Fig. 1 on a stereographic projection, full and open circles representing north poles on the lower and upper hemispheres respectively. All the rocks collected by Wheeler and Green are reversely magnetized while the rest are normal except for two of Poole's samples, which are almost horizontally polarized.

The mean direction of these rock samples has been computed by reversing all the reversely polarized rocks and adding them to the normal ones. The two horizontally polarized samples are considered to be anomalous, due perhaps to magnetization caused by lightning, and have not been included in the statistical analysis. The mean direction, H, with its Fisher statistics, has been listed in Table 1, where it is compared with the field, F, produced by a central axial dipole and the present geomagnetic field, F', at

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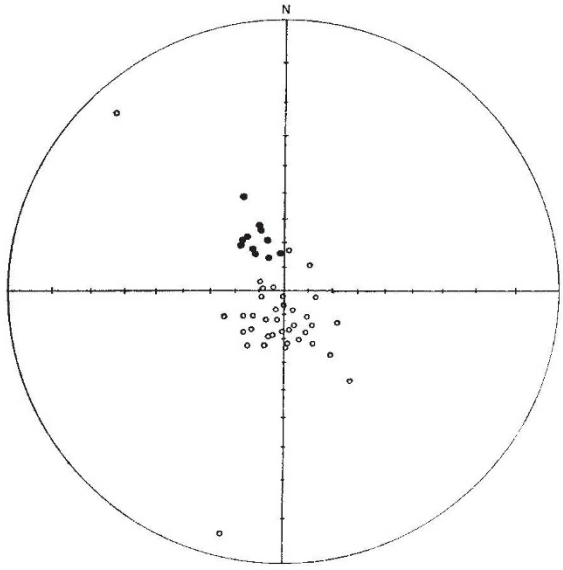


Fig. 1. Stereographic projection of palæomagnetic directions of Late Tertiary volcanics from north-western Canada. ●, North pole on lower hemisphere; ○, north pole on upper hemisphere

60·0° N. and 135·0° W. The λ's are the angles between the palæomagnetic direction, H, and F and F'.

Table 1 shows that the mean palæomagnetic direction of these Late Tertiary rocks is equivalent to the axial dipole field expected at this locality and is significantly different from the Earth's present field. These results are therefore further evidence to support the basic assumption of most geophysical interpretations of palæomagnetism, namely, the average magnetic field of the Earth is due to a geocentric axial dipole.

Table 1

	H	F	F'
D	348·5	0·0°	31·0°
I	+ 75·0	+ 74·0°	+ 76·0°
θ	4·0°	—	—
K	28·0	—	—
N	46	—	—
λ	0°	3·5°	10·5°

D, Declination east of north; I, inclination (+ north pole down); θ, angular radius of circle of confidence at 95 per cent level; K, Fisher's dispersion; N, number of samples.

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P. M. Du Bois

Geological Survey of Canada,
Ottawa. Feb. 23.

PHYSICAL SCIENCES

Simultaneous Occurrence of Sub-Visual Auroræ and Radio Noise Bursts on 4·6 kc./s.

Since July 1958 the red oxygen (6300 A.) airglow has been recorded at Camden, near Sydney (geomagnetic latitude 42° S.), with a sky-scanning photometer. This instrument has been described by St. Amand[1]. These observations frequently detected greatly enhanced emission towards the south, and less often structures such as arcs and rays. We interpret these as auroræ, usually sub-visual, though on seven nights auroræ were seen.

Table 1. Occurrence and Absence of Noise on 4·6 kc./s. and Auroræ

		Noise on 4·6 kc./s.	
		Yes	No
Auroræ	Yes	18	2
	No	6	27

During the same period, radio noise on 4·6 kc./s. was continuously recorded by a method[2] which largely eliminates impulsive atmospheric and industrial interference. These observations[3] frequently detected distinct noise bursts, many times stronger than the background and usually lasting a few hours.

A comparison of the radio and nightglow records shows that there is a definite correlation between the two. Of 53 nights of simultaneous observations of noise and airglow between June and December 1958, noise bursts and sub-auroræ were recorded during the same hours on 18 nights. On 27 nights neither noise nor sub-auroræ were recorded; on 6 nights noise was recorded in the absence of any auroral pattern in the airglow, while on two occasions auroræ were recorded without any noise bursts. This is summarized in Table 1.

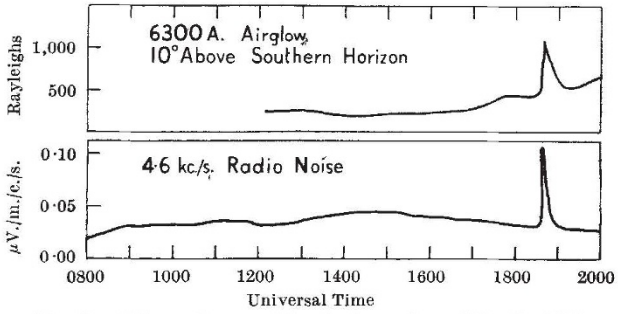


Fig. 1. Noise and auroral burst observed on July 21, 1958

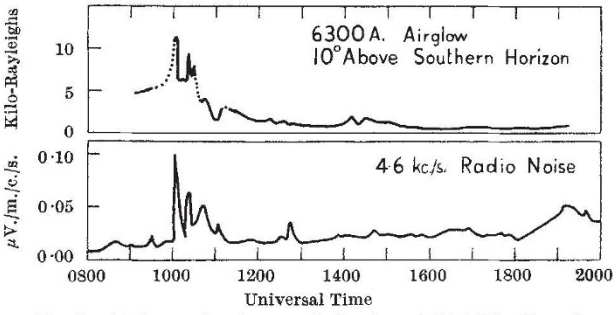


Fig. 2. Airglow and noise records for August 27, 1958. No red airglow observations were made during those periods for which the trace is dotted

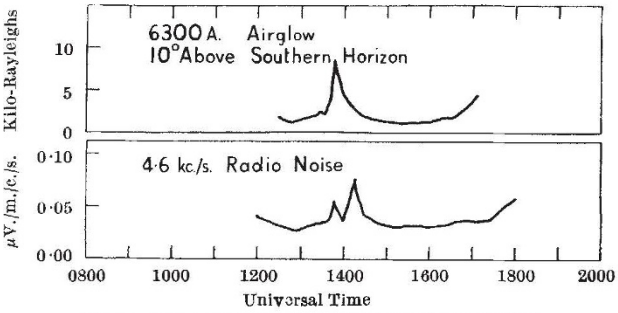


Fig. 3. Noise and airglow records for January 5, 1959

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